then, since we do not really understand at all well the laws of quantum gravity and their consequences.

This aging of a black hole's internal tidal forces is not inexorable. Whenever matter and radiation (or astronauts) fall into the hole, they will feed and energize the tidal forces, much like a hunk of meat thrown to a lion energizes him. The oscillatory stretch and squeeze near the singularity, having been fed, will grow stronger for a short while, and then will die out and become quiescent once again.

In the late 1950s and early 1960s John Wheeler had a dream, a hope, that we humans might one day be able to probe into a singularity and there see quantum gravity at work—that we might probe not only with mathematics and computer simulations, but also with real, physical observations and experiments. Oppenheimer and Snyder dashed that hope (Chapter 6). The horizon that they discovered forming around an imploding star hides the singularity from external view. If we remain forever outside the horizon, there is no way that we can probe the singularity. And if we plunge through the horizon of a huge old hole, and survive to meet the quantum gravity singularity face-to-face, there is no way we can transmit a description of our meeting back to Earth. Our transmission cannot escape from the hole; the horizon hides it.

Though Wheeler has long since renounced his dream and now vigorously champions the view that it is impossible to probe singularities, it is not at all certain that he is correct. It is conceivable that some extremely nonspherical stellar implosions produce *naked singularities*, that is, singularities that are not surrounded by horizons and that therefore can be observed and probed from the external Universe, even from Earth.

In the late 1960s, Roger Penrose searched hard, mathematically, for an example of an implosion that creates a naked singularity. His search came up empty. Whenever, in his equations, an implosion created a singularity, it also created a horizon around the singularity. Penrose was not surprised. After all, if a naked singularity were to form, then it seems reasonable to expect that, just before the singularity forms, light can escape from its vicinity; and if light can escape, then (it would seem) so can the material that is imploding to create the singularity; and if the imploding material can escape, then presumably the material's huge internal pressure will make it escape, thereby reversing the implosion and preventing the singularity from